

Theoretical exploration of Orth-C₁₄ as a potential candidate for the anode material of Li-ion batteries

Han Zhang^{a,†}, Zilin Chen^{a,†}, Huimin He^a, Qiuyan Lu^a, Huadong Zeng^{a,*} , Xinlu Cheng^b

^a School of Materials and New Energy, South China Normal University, Shanwei 516625, China

^b Institute of Atomic and Molecular Physics, Sichuan University, Chengdu 610065, China

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ABSTRACT

The exploration of two-dimensional (2D) carbon allotropes as high-performance anode materials in Li-ion batteries (LIBs) has attracted experimental and theoretical research interest. Inspired by this, based on first-principles calculations, we comprehensively evaluate the viability of a novel 2D Orth-C₁₄ monolayer regarded as a prospective anchoring material for LIBs. The multifaceted analyses are conducted to unravel the stability and metallicity of the Orth-C₁₄ monolayer. The calculations indicate that the Orth-C₁₄ monolayer possesses the preferable anchoring strength (-1.78 eV) and charge transfer (0.97 e) toward lithium. To meet the requirement of an excellent anode material, the theoretical Li storage capacity of the Orth-C₁₄ monolayer reaches up to 956 mAh/g. More notably, the Orth-C₁₄ monolayer exhibits an extremely low diffusion barrier (0.12 eV), and maintains suitable open circuit voltage (0.053 – 0.393 V). The Orth-C₁₄ monolayer displays exceptional structural stability with only 1.3 % volume expansion during cycling, contributing to its outstanding cycle life. The potential of the Orth-C₁₆ monolayer, a structural analog of Orth-C₁₄, as a LIB anode is further analyzed. Finally, the intrinsic metallicity, low diffusion barrier, high capacity, excellent cycling performance of the Orth-C₁₄ monolayer position it as a promising candidate for advanced LIBs applications.

1. Introduction

Among the alkali metal, Li-ion batteries (LIBs) have attracted substantial attention in the energy storage device industry due to their high energy density and long lifespan, leading to their widespread application in mobile electronic devices and smart power networks [1–6]. The anode material constitutes the critical determinant of battery performance metrics. However, it has been shown that Li-ion intercalation in graphite occurs exclusively through edge-plane sites, and the resultant sluggish diffusion kinetics lead to significant capacity fading. The limited theoretical storage capacity (372 mAh/g) and suboptimal interlayer Li diffusion kinetics are nearing their theoretical maximum, failing to meet the escalating demands for high energy density and fast-charging capabilities in electric vehicles and aerospace applications.

To address the challenge of energy density limitations of LIBs, the development of two-dimensional (2D) anode materials represents a promising strategy. The atomic-scale thickness of two-dimensional materials significantly minimizes Li-ion diffusion distances and greatly enhances diffusion rates, and their low diffusion barriers and large

surface area collectively demonstrate extraordinary physicochemical properties. Owing to their excellent electrical conductivity and chemical stability, 2D carbon-based materials hold a pivotal role [7,8].

The novel 2D carbon allotrope has garnered significant attention as a research hotspot in the academic community. Researchers have successively designed and proposed a series of novel 2D carbon allotropes featuring distinctive pore structures and electronic properties. The 2D carbon allotropes can form heterojunctions with other materials through physical stacking or chemical conjugation, facilitating synergistic optimization of performance. Compared to individual components, the integrated C₄N/graphene system not only manifests a significant improvement in conductivity, but also has a high specific capacity [9]. Moreover, 2D carbon allotropes can be applied to stress manipulation to obtain target properties. For example, the best performance can be obtained by altering the adsorption and mechanical properties of materials under biaxial tensile strain [10]. Their large specific surface area provides abundant exposed active sites, which not only increases energy density but also mitigates the capacity limitations of bulk materials [11–15]. Among these alternatives, strategically designed 2D carbon

* Corresponding author.

E-mail address: zenghd@m.scnu.edu.cn (H. Zeng).

† These authors contributed equally to this work.

allotropes, such as graphdiyne, T-carbon and penta-graphene offer a novel strategy to transcend the performance limitations of conventional carbon materials through innovative atomic configurations [16–18]. For example, Petal-Graphyne, consisting of 4-, 8-, 10- and 16-membered carbon rings, exhibits a high theoretical capacity of 1004 mAh/g, a low diffusion barrier for Li-ions (0.28 eV), and an average open circuit voltage of 0.47 V [19]. Net-C18 has a low energy barrier (0.32 eV) and a theoretical specific capacity of 403 mAh/g [20]. While 2D carbon allotropes show great potential as Li-ion battery anodes, they typically struggle to balance high capacity with low diffusion barriers. Consequently, studying new 2D carbon allotropes with both high energy density and excellent rate performance becomes the focus of current research.

The recent reporting of a novel 2D carbon allotropes, Orth-C₁₄, has attracted considerable [21]. Studies find that the electrical properties of these materials can be changed through strain engineering and they exhibit extremely high electron mobility. However, there are almost no research reports on the Orth-C₁₄ monolayer as an anode material for LIBs, and its intrinsic physical mechanisms have not yet been fully explained from the perspective of theoretical calculations. The first-principles density functional theory (DFT) calculations are employed to systematically investigate the practical feasibility of the Orth-C₁₄ monolayer as an anode for LIBs. Initially, the Orth-C₁₄ monolayer model is constructed and its stability and electronic properties are examined, revealing excellent stability and metallic characteristics. Subsequently, we investigate the adsorption of Li-ions on the Orth-C₁₄ monolayer by progressively binding Li atoms into the system, demonstrating stable adsorption behavior. The study then proceeds to quantify its electrochemical performance, with the Orth-C₁₄ monolayer displaying an ideal combination of properties: low diffusion barrier, low volume expansion rate, suitable open circuit voltage (OCV) and exceptional theoretical storage capacity. Hence, the Orth-C₁₄ monolayer displays excellent potential as an advanced anode for LIBs.

2. Computational methods

All first-principles calculations are performed utilizing the Vienna *Ab-initio* Simulation Package (VASP, version 5.4.4) based on density functional theory (DFT) and pseudopotential methods [22,23]. The interaction between the core and valence electrons is taken into account using the projector augmented wave (PAW) approach with a plane-wave basis [24]. The generalized gradient approximation (GGA) in the framework of the Perdew-Burke-Ernzerhof (PBE) functional is adopted to describe exchange-correlation effects [25]. Besides, the semi-empirical method of Grimme (DFT-D3) correction is also employed to estimate the van der Waals (vdW) interactions [26,27]. A vacuum layer of 20 Å is introduced perpendicular to the surface to effectively eliminate interlayer interactions. All related computations utilize a plane-wave cutoff energy of 550 eV. For structural relaxation and density of states (DOS) calculations, k-point grids of $4 \times 4 \times 1$ and $8 \times 8 \times 1$ are implemented, respectively, to balance computational efficiency and accuracy. The structural relaxation achieves full optimization of atomic positions and lattice constants when atomic forces fall below 0.01 eV/Å, while self-consistent field (SCF) convergence is considered to be converged upon reaching a total energy variation threshold of 10^{-6} eV. The phonon dispersion is calculated via the density functional perturbation theory (DFPT) [28]. The VASPKIT toolkit is employed for systematic data post-processing and analysis [29]. The *ab-initio* molecular dynamics (AIMD) simulations are performed under the canonical ensemble (NVT) at 500 K for 5000 fs, with a time step of 1 fs [30]. A $2 \times 2 \times 1$ supercell (56 atoms) is constructed to investigate the adsorption and diffusion behaviors. Furthermore, the climbing-image nudged elastic band (CI-NEB) method is employed to identify Li-ion diffusion pathways and calculate the energy barrier within the Orth-C₁₄ monolayer [27]. The charge distribution and transfer are estimated by the Bader charge analysis.

3. Results and discussion

3.1. Crystal structure, stability and electronic properties of Orth-C₁₄ monolayer

The Orth-C₁₄ monolayer, a planar sp²-hybridized 2D carbon allotrope in the *Pmm2* space group, exhibits a mixed bonding configuration with coexisting C–C single bonds and C = C double bonds. Its framework primarily comprises four-membered, six-membered, and eight-membered rings. Fig. 1 illustrates the Orth-C₁₄ monolayer from top and side perspectives. We label four non-equivalent carbon atoms (C1 – C4) at their Wyckoff positions. The optimized geometric structure of the Orth-C₁₄ monolayer demonstrates lattice constants of $a = 8.750$ Å and $b = 4.351$ Å, displaying a deviation by only 0.4 % from theoretical data. Meanwhile, the C–C and C = C bond lengths in the Orth-C₁₄ crystal structure range from 1.39 to 1.52 Å and 1.40 to 1.46 Å, respectively, and both are in excellent agreement with previously reported values [21]. Successful experiments in fabricating ordered 2D carbon allotropes have paved a promising path toward the eventual synthesis of Orth-C₁₄ [31, 32]. They report a method for synthesizing non-benzenoid carbon allotropes through sequential Ullmann coupling, dehydrogenative cyclization, and final HF-zipping of 4,4"-dibromo-2,2',2'',5,5',5''-hexafluoro-1,1':4',1''-terphenyl (DHTP) precursor on Au (111) surface.

For evaluating the thermodynamic and dynamical stability of the Orth-C₁₄ monolayer, we perform calculations of the phonon spectra and AIMD simulations. The phonon spectra of the Orth-C₁₄ monolayer are presented in Fig. 2(a). The absence of imaginary frequencies across the entire Brillouin zone confirms the dynamic stability of this structure. It is observed that the maximum frequency in the phonon spectra of the Orth-C₁₄ monolayer (56.72 THz) exceeds that of graphene (49.50 THz) [33]. Then, the AIMD simulations are employed to evaluate the thermal stability of the Orth-C₁₄ monolayer for 5 ps in a $2 \times 2 \times 1$ supercell at 500 K and 1000 K. Fig. 2(b) demonstrates that the Orth-C₁₄ monolayer maintains its structural integrity with minimal potential energy fluctuations, confirming thermal stability. During the specified simulation period, no C–C bond breaking or ring structure distortion is observed in either of the two structures. Our results further indicate that the Orth-C₁₄ monolayer manifests exceptional thermal stability, maintaining complete structural integrity at elevated temperatures.

To study the electronic properties of the Orth-C₁₄ monolayer, the band structure, DOS and electron localization function (ELF) are calculated. The electronic band structure and corresponding DOS of the Orth-C₁₄ monolayer are illustrated in Fig. 3(b). The direct overlap between the valence and conduction bands, along with two partially filled bands crossing the Fermi level, reveals that the Orth-C₁₄ monolayer is metallic and conducts electricity. Furthermore, partial density of states (PDOS) of three non-equivalent carbon atoms is calculated as shown in Fig. 3(c). Analysis reveals that the electronic states near the Fermi level in the Orth-C₁₄ monolayer are primarily contributed by C-2p orbitals. The ELF for the Orth-C₁₄ monolayer is plotted in Fig. 3(a). According to the definition of previous research [34], the ELF can be visualized as a contour map in real space, which is normalized within the range 0 to 1. The analysis results demonstrate that the ELF values of C–C bonds are ~ 0.75 in the Orth-C₁₄ monolayer, indicating that strong covalent bonds are formed between carbon atoms through electron sharing. These research findings indicate the excellent electrical conductivity of 2D carbon materials. The unique structure, good stability, and inherent metallic properties of the Orth-C₁₄ monolayer suggest its potential as an anode material for LIBs.

3.2. Adsorption and diffusion of single Li-ion on Orth-C₁₄ monolayer

The stable adsorption capability of Li atoms on the Orth-C₁₄ monolayer surface constitutes a fundamental prerequisite for its feasibility as an anode material for LIBs. The Li-ion adsorption calculations are conducted on the $2 \times 2 \times 1$ supercell of the Orth-C₁₄ monolayer.

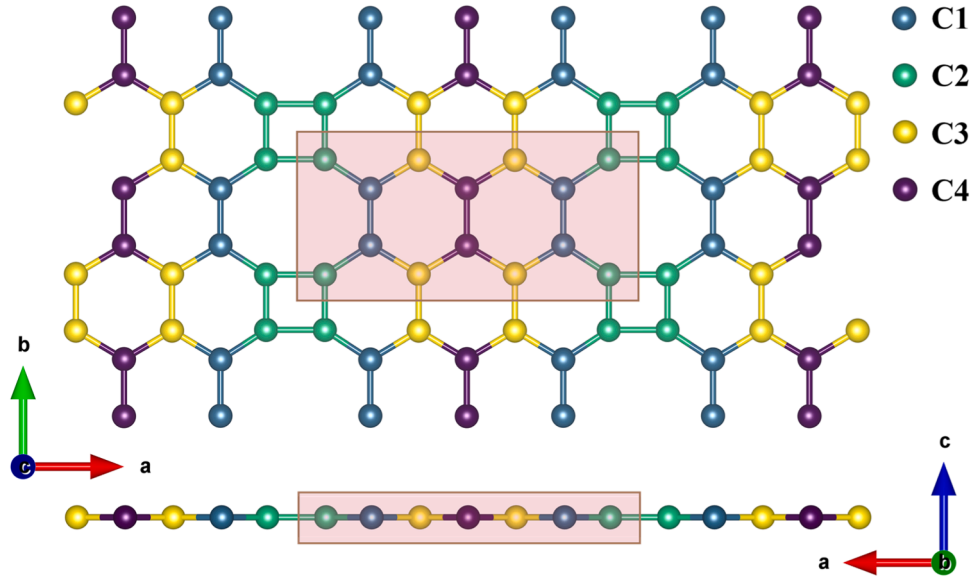


Fig. 1. The top and side view of the Orth- C_{14} monolayer, different colors in the picture represent unequal carbon atoms.

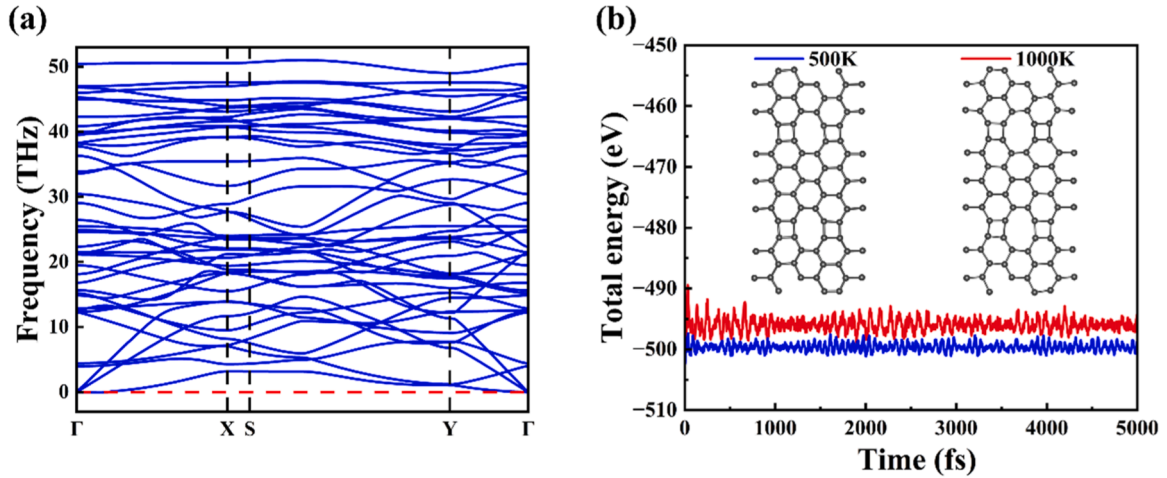


Fig. 2. (a) The calculated phonon spectra of the Orth- C_{14} monolayer using a $2 \times 2 \times 1$ supercell. (b) The variation of the total energy of Orth- C_{14} monolayer simulated by AIMD at 500 K and 1000 K and the insets show the final snapshots, respectively.

Considering the symmetrically equivalent sites, the eleven potential adsorption sites are designed on the supercell: three top sites (T1, T2, and T3), five bridge sites (B1, B2, B3, B4, and B5), and three hollow sites (H1, H2, and H3). After complete structural relaxation, both the top and bridge sites spontaneously migrate to hollow sites. (see Fig. S1) This phenomenon signifies that the hollow sites are the most stable adsorption sites for Li atoms on the Orth- C_{14} monolayer, and the structural distortion during the process can be ignored. Consequently, all subsequent analyses focus solely on hollow site configurations. The adsorption energy of Li at different H sites on the Orth- C_{14} monolayer is calculated as follows:

$$E_{ad} = E_{Li-Orth-C_{14}} - E_{Orth-C_{14}} - E_{Li} \quad (1)$$

where E_{ad} is the adsorption energy for a single Li atom on the Orth- C_{14} monolayer, $E_{Li-Orth-C_{14}}$ and $E_{Orth-C_{14}}$ are the total energies of the Orth- C_{14} monolayer with and without one Li adatom, respectively and E_{Li} is the energy per atom in the bulk Li metal. As shown by the computational results (Fig. 4(a)), the presence of negative values indicates that the process is thermodynamically exothermic and spontaneous. The more negative the adsorption energy, the greater the stability of the system

due to stronger binding interactions. Our calculations identify the H1 site as the most stable adsorption configuration with the lowest adsorption energy (-1.78 eV), while the H2 (-1.71 eV) and H3 (-1.74 eV) represent metastable states in descending order of stability. The H1 site occupies the central position within the octagonal geometry, where the larger internal angles create an expanded cavity that optimally accommodates a Li atom, ensuring exceptional adsorption stability.

To visually probe the electron transfer during adsorption of a single Li atom on the Orth- C_{14} monolayer, we further analyze the charge density difference (CDD) for the Li atom at the H1 site. The calculation formula is as follows [35,36]:

$$\Delta\rho = \rho_{Orth-C_{14}+Li} - \rho_{Orth-C_{14}} - \rho_{Li} \quad (2)$$

where $\rho_{Orth-C_{14}+Li}$, $\rho_{Orth-C_{14}}$, and ρ_{Li} are the charge densities of the total adsorption structure, pristine Orth- C_{14} monolayer, and one Li atom adsorption on Orth- C_{14} . The electron density gradient difference in Fig. 4(b) indicates a charge transfer from the Li atom to the Orth- C_{14} monolayer, where yellow and cyan regions represent electron accumulation and depletion respectively. A large number of electrons accumulate above and below the Orth- C_{14} monolayer, while electron density

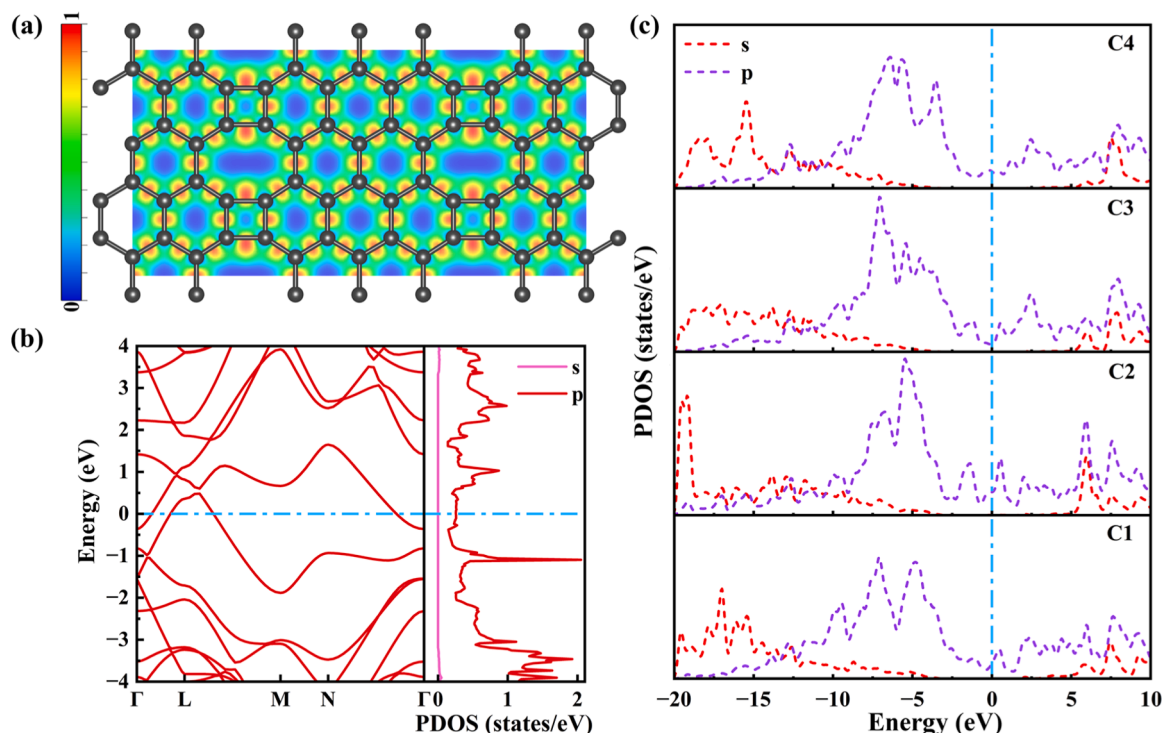


Fig. 3. (a) The ELF map, (b) band structure and PDOS of the Orth- C_{14} monolayer. (c) The PDOS of four non-equivalent carbon atoms of the Orth- C_{14} monolayer.

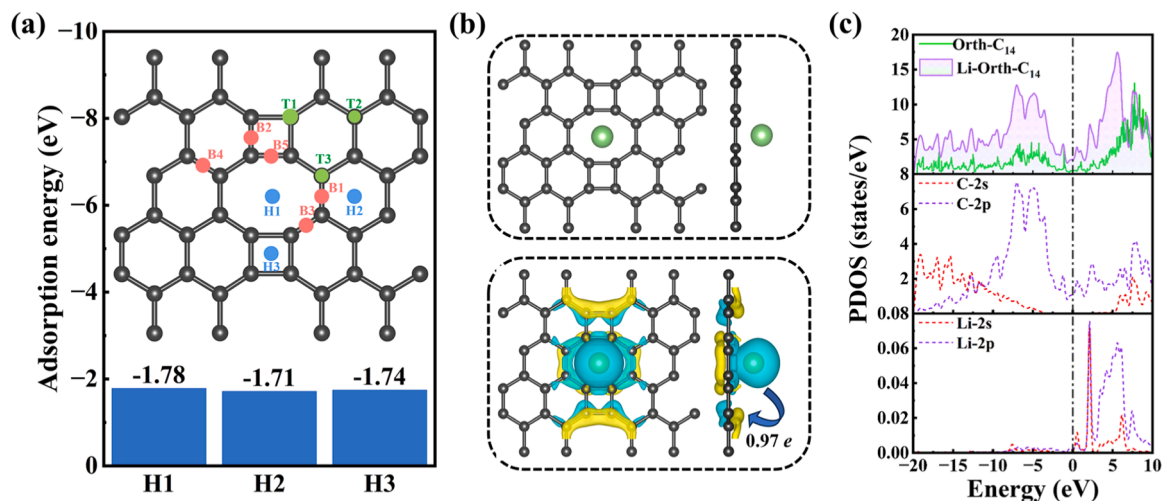


Fig. 4. (a) The possible adsorption sites and corresponding adsorption energy of Li atom on the Orth- C_{14} monolayer. (b) The optimized structure of single Li atom adsorbed on the Orth- C_{14} monolayer at H1 site, and the CDD for the Li atom adsorption on the Orth- C_{14} monolayer. (c) The DOS and PDOS of Orth- C_{14} monolayer and Li-adsorbed Orth- C_{14} monolayer.

decreases near Li atoms. The electron transfer from Li to the substrate leads to Li ionization, with the resulting Li-ions becoming firmly anchored to the surface of the Orth- C_{14} monolayer. Through Bader charge analysis, the charge transfer in the adsorption system is quantified, indicating that nearly a full electron ($0.97 e$) is transferred from the Li atom to the Orth- C_{14} monolayer. This significant charge transfer indicates the presence of strong electrostatic interactions between Li-ion and the Orth- C_{14} monolayer. The DOS and PDOS for the Orth- C_{14} monolayer, with and without an adsorbed Li atom, are calculated to probe their interaction and are presented in Fig. 4(c). A significant expansion of the DOS curve area is observed at the Fermi level following adsorption, indicating that their interaction enhances electronic states near the Fermi level, which suggests improved electrical conductivity.

This enhancement directly benefits key battery performance, supporting the feasibility of Orth- C_{14} as an anode for LIBs.

Fast charging and discharging rates are key attributes of high-quality electrode materials, directly affecting the performance and application efficiency of batteries. The effectiveness of these rates primarily depends on the diffusion rate of Li-ions on the substrate. Therefore, materials with high Li-ion diffusion rates are crucial for improving the overall performance of batteries. To investigate Li-ion diffusion on the surface of the Orth- C_{14} monolayer, three possible diffusion pathways are designed at the stable H sites on the Orth- C_{14} monolayer. Fig. 5 displays the diffusion barriers for Li-ions on the Orth- C_{14} monolayer. Path 1 ($H1 \rightarrow H2$) exhibits a diffusion barrier of $0.27 eV$, Path 2 ($H1 \rightarrow H3$) shows a barrier of $0.23 eV$, while Path 3 ($H3 \rightarrow H4$) demonstrates a significantly

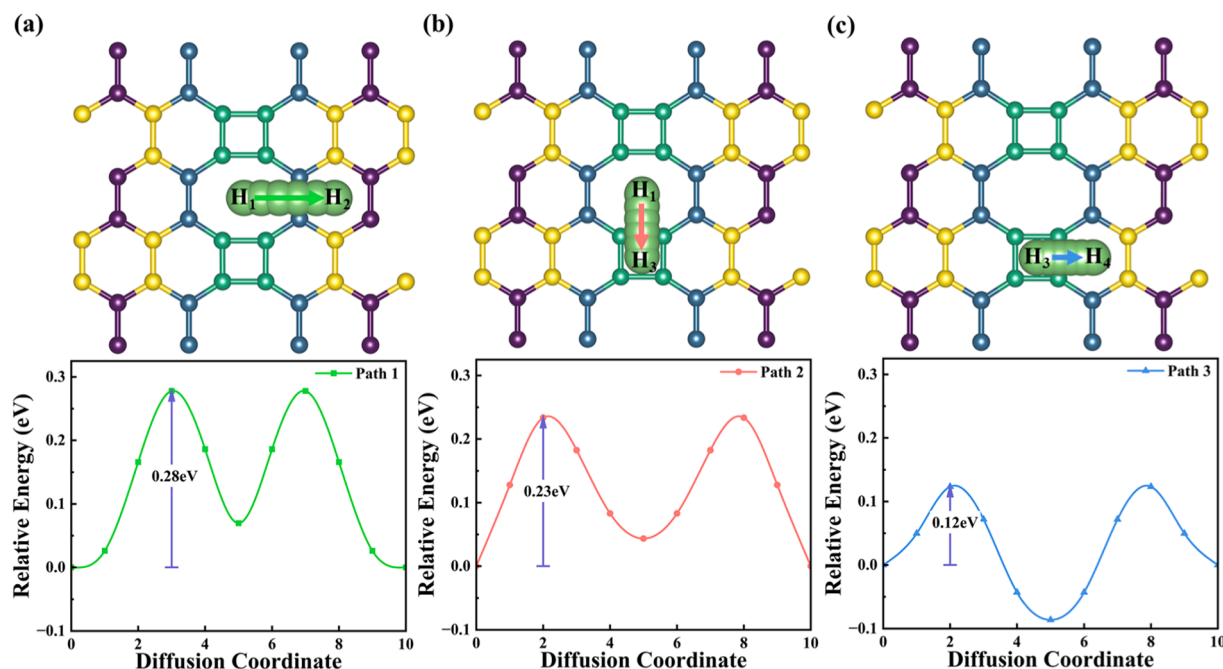


Fig. 5. (a-c) The schematic diagram of the diffusion pathway in the Orth- C_{14} monolayer, the corresponding energy barrier profiles of three diffusion paths on images.

lower barrier of 0.12 eV. With an ultra-low diffusion energy barrier of 0.12 eV, the Orth- C_{14} monolayer reveals substantially faster Li-ion transport kinetics than novel 2D electrode materials, including TODD-graphene (0.83 eV), holey graphyne (0.28 eV), and so forth [18, 37–43]. A comprehensive comparison of these diffusion barriers is provided in Table 1. The minimal Li-ion diffusion barrier observed on the Orth- C_{14} monolayer points to superior rate performance and efficient charge-discharge capabilities, which are essential traits for high-power LIB anode materials.

Table 1

Comparison of Orth- C_{14} with other carbon-based materials in terms of diffusion barrier (eV), theoretical storage capacity (mAh/g) and OCV (V).

Materials	Diffusion barrier	Theoretical storage capacity	OCV
graphite [46]	0.22–0.40	372	0.11
PAI-graphene [53]	0.34	744	0.190–0.324
Θ-graphene [57]	0.45–0.48	876.65	0.30
xgraphene [48]	0.44–0.49	930	0.28
bco- C_{16} [55]	0.53–2.23	558	0.23
penta-graphene [18]	0.04–0.17	1489	0.50–0.60
LOPC- C_{52} [54]	0.38–0.52	1244	0.20
M-graphene [47]	0.34–0.37	279	0.548
net- C_{18} [20]	0.32–0.72	403	0.024–1.27
thgraphene [51]	0.06–0.17	744	0.30
popgraphene [39]	0.55	1487	0.45
C_{568} [40]	0.42	1373	/
B-graphyne [50]	0.17–0.87	525	0.20–1.23
net- τ [52]	0.30–0.36	558	0.039–0.608
ZW-21 [49]	0.46–0.50	634	/
holey graphyne [58]	0.28	873	/
h-AIC [7]	0.78–0.85	739.61	1.38
α -CP [8]	0.11	1108.91	1.99
Petal-Graphyne [19]	0.28	1004	0.47
TPDH-graphene [59]	0.08–0.20	1116	0.29
Orth- C_{14}	0.12–0.28	956	0.053–0.393

3.3. Storage capacity and OCV of Orth- C_{14} monolayer

The storage capacity, a critical anode performance metric, is fundamentally determined by the maximum adsorbable concentration of Li atoms within the substrate material. It is worth noting that the theoretical maximum storage capacity of the Orth- C_{14} monolayer is strongly dependent on Li adsorption density. To evaluate the theoretical maximum Li storage capacity of the Orth- C_{14} monolayer, we gradually add Li atoms on both sides. Avoiding interactions between adjacent Li-ions caused by tiny interatomic distances, we employ the $2 \times 2 \times 1$ supercell of the Orth- C_{14} monolayer as the substrate throughout this study. Specifically, the average adsorption energy of the stepwise adsorption of multiple Li atoms at the Orth- C_{14} monolayer is calculated, which can be defined as the following equation:

$$E_{ave} = \frac{E_{nLi+Orth-C_{14}} - nE_{Li} - E_{Orth-C_{14}}}{n} \quad (3)$$

In the formula, E_{ave} represents the average adsorption energy per Li atom on the Orth- C_{14} monolayer and n is the total number of adsorbed Li atoms. Here, $E_{nLi+Orth-C_{14}}$ and $E_{Orth-C_{14}}$ are the total energies of the system with and without n Li-ions adsorbed, respectively, and E_{Li} represents the energy of one Li atom in bulk Li metal. The negative adsorption energy indicates stable Li atom adsorption on the Orth- C_{14} monolayer, which effectively avoids clustering effects. Given that the H1 site has superior stability, we prioritize Li adsorption at all H1 sites before populating other H sites. The decreases with the increasing number of Li atoms, owing to the repulsion forces of neighboring Li atoms [44]. Fig. 6(b) shows the average adsorption energies for Li concentrations of $n = 4, 8, 10, 12$, and 24, with values of -1.573 eV, -1.423 eV, -1.361 eV, -1.314 eV and -1.279 eV, respectively. The Li atoms are incrementally adsorbed on the Orth- C_{14} monolayer until geometric optimization fails to converge, with all stable H sites occupied by Li atoms, thus determining a saturation limit of 12 Li atoms per layer. Similarly, we also gradually add 12 Li atoms to the opposite side of the monolayer. The adsorption energies demonstrate a decreasing trend with the number of Li atoms increasing. However, extrapolation of this trend would suggest persistently negative adsorption energies without transitioning to positive values. The positive adsorption energy is not a prerequisite for determining adsorption completion [45]. The

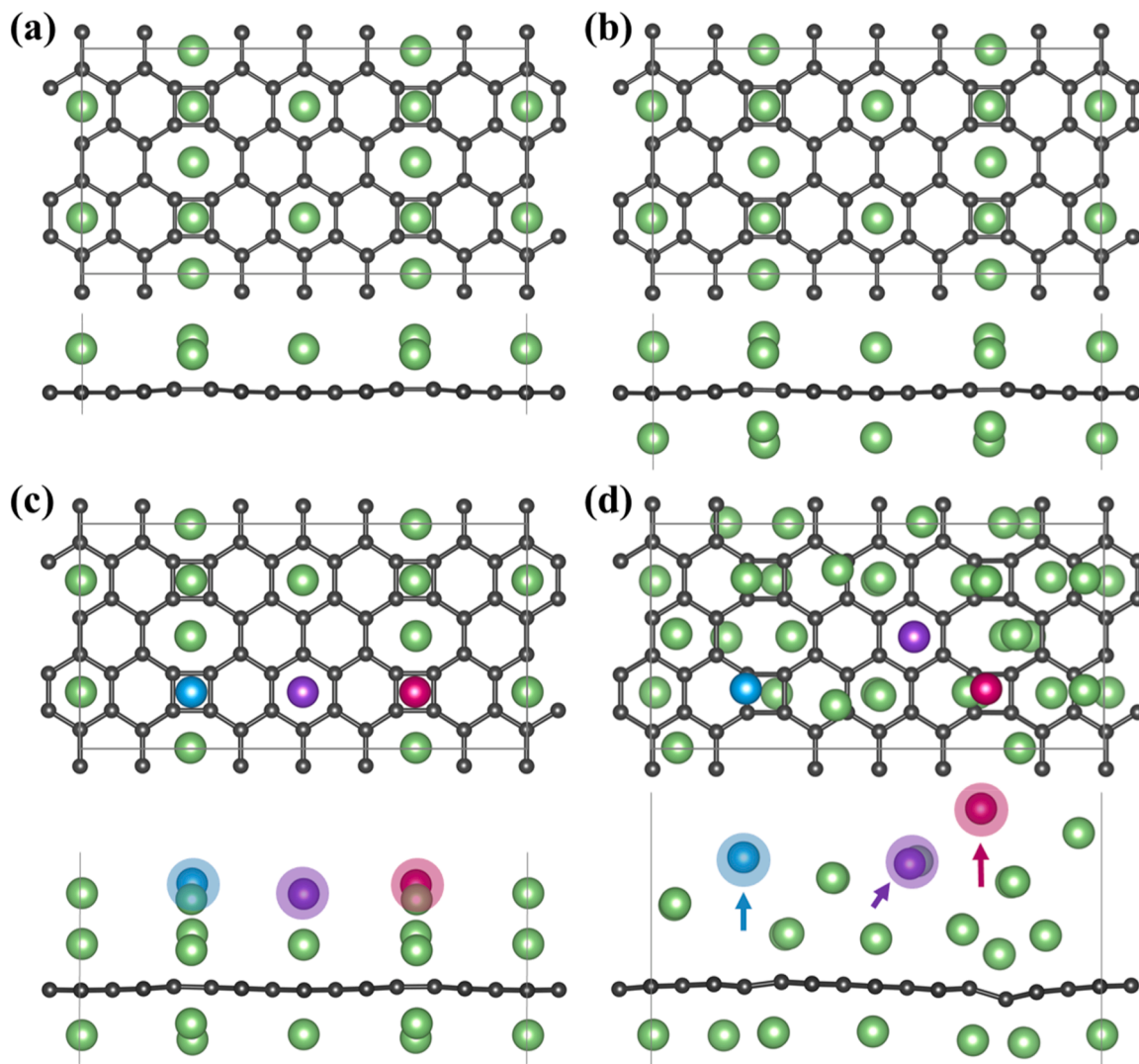


Fig. 6. (a-b) The top and side view for Li-adsorbed Orth-C₁₄ monolayer with monolayer and bilayer Li-ion. (c-d) The top and side view of a three-layer Li-ion adsorption Orth-C₁₄ monolayer before and after geometric relaxation.

adsorption ceases when the host material reaches saturation and can no longer accommodate additional atoms. The structural relaxation of the three-layer Li-ion adsorption model fails to achieve convergence, the third layer Li atoms migrating from their initial positions, leading to dendrite formation. In addition, we find that Orth-C₁₄ undergoes distortion after structural relaxation (see Fig. 6(c-d)).

Computational results reveal that the Orth-C₁₄ monolayer can stably accommodate up to 24 Li atoms, corresponding to a full monolayer coverage of 12 atoms on each side (see Fig. 7(a-b)). The most stable adsorption configuration for Li on the Orth-C₁₄ monolayer is determined by evaluating all possible adsorption sites and selecting the one with the lowest energy. The theoretical storage capacity of a system is predicted by using the following equation:

$$C_M = \frac{n \times F}{M_{Li_{max}Orth-C_{14}}} \quad (4)$$

In the formula, C_M is the valence of the adsorbed Li atoms, n is the number of adsorbed Li atoms, F is the Faraday constant (26,801 mA h/mol), and $M_{Li_{max}Orth-C_{14}}$ represents the molecular weight of the supercell. Hence, the theoretical specific storage capacity is calculated to be 956 mAh/g. This value compares favorably with, and in many cases exceeds, those of other reported 2D carbon allotropes, such as M-graphene (279 mAh/g), graphite (372 mAh/g), xgraphene (930 mAh/g), popgraphene

(1487 mAh/g), C₅₆₈ (1373 mAh/g) and so on [39,40,46–51]. (Table.1) The Orth-C₁₄ monolayer has great potential for achieving higher capacity, and its theoretical storage capacity surpasses most reported two-dimensional carbon materials.

OCV is the voltage when the battery is not connected to a load, which can reflect the state of charge and state of health, serving as an important metric for LIBs. The OCV for the Orth-C₁₄ monolayer at incremental Li atoms adsorption stages is determined, with the results presented in Fig. 6(a). For the charge process simulation, the OCV potential is determined through the following theoretical formulation:

$$OCV = \frac{E_{Orth-C_{14}} + xE_{Li} - E_{Li_x+Orth-C_{14}}}{xe} \quad (5)$$

Here $E_{Li_x+Orth-C_{14}}$ denotes the total free energy of the system with x Li atoms adsorbed, e is the charge constant, and x is the number of adsorbed Li atoms, and other parameters are in accordance with the previous equations. A higher OCV increases the risk of lithium dendrite formation and may lead to short circuits. To ensure battery safety, it is critical to maintain an optimal OCV range. Our calculations show that OCV exhibits a monotonic decrease with increasing Li concentration. The OCV of the Orth-C₁₄ monolayer is computationally determined for x Li atoms adsorbed per unit cell. It exhibits a concentration-dependent decrease, with values of 0.393 V ($x = 4$), 0.178 V ($x = 8$), 0.136 V (x

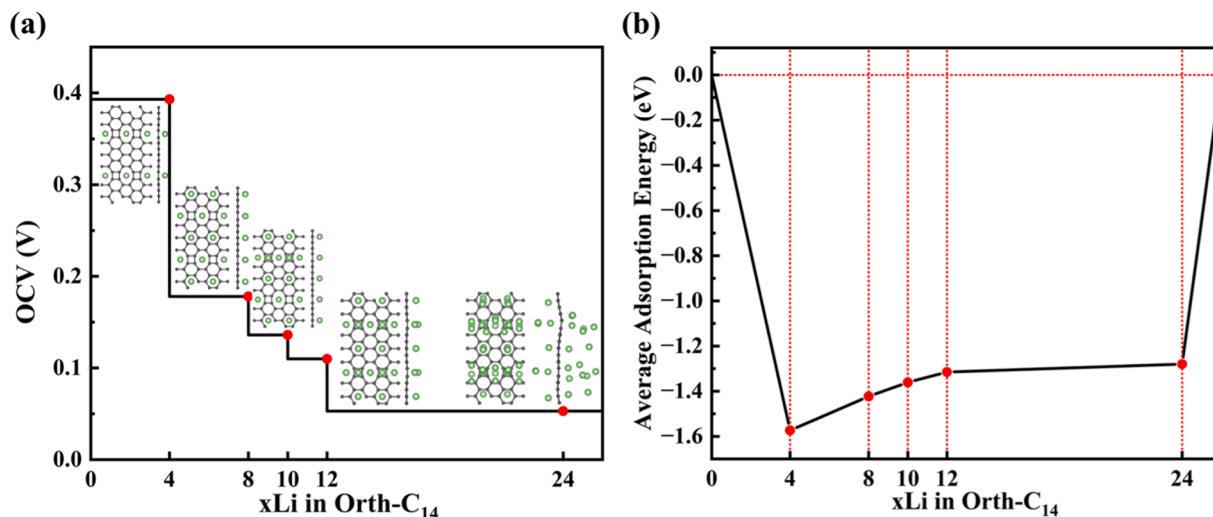


Fig. 7. (a) The OCV profile and (b) the average adsorption of the Orth-C₁₄ monolayer with 4, 8, 10, 12, 24Li atoms concentrated.

= 10), 0.110 V ($x = 12$) and 0.053 V ($x = 24$). The range of OCV after the gradual adsorption of Li atoms by the Orth-C₁₄ monolayer is relatively moderate in terms of 2D carbon materials, such as net-C18 (0.024 – 1.27 eV), net- τ (0.039 – 0.608 eV), PAI-graphene (0.019 – 0.324 eV) [20,52, 53]. The moderate OCV maintained by the Orth-C₁₄ monolayer throughout the charging process enables good electrochemical

performance and optimal operational voltage characteristics in LIBs.

To comprehensively assess its thermal stability, AIMD calculations are allowed for further investigation of this aspect. Fig. 8(a-b) presents the AIMD simulation of the Orth-C₁₄ monolayer with 24 adsorbed Li atoms at 500 K for 5 ps, with a time step of 1 fs. The results suggest only minor structural perturbations with no evidence of bond breaking or

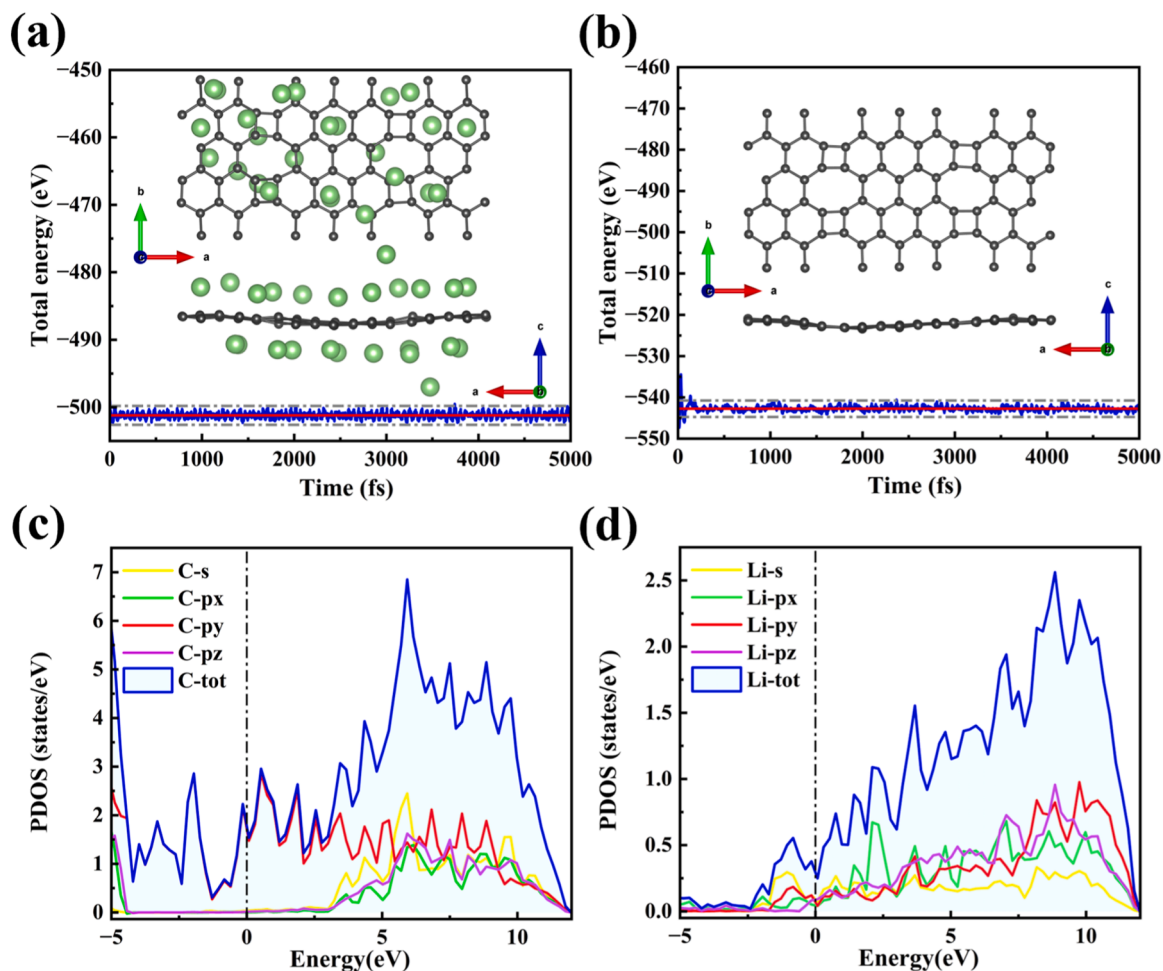


Fig. 8. The fluctuation of total energy during the AIMD simulations of (a) the adsorption of 24 Li atoms and (b) the desorption of 24 Li atoms on Orth-C₁₄ at 500 K. The PDOS of substrate elements (c) C, (d) metal Li in Orth-C₁₄ monolayer adsorbed by bilayer Li atoms.

lattice reconstruction. The Orth-C₁₄ monolayer exhibits excellent thermal stability and is expected to be structurally robust at room temperature after Li adsorption. To simulate the actual battery charging and discharging process, all the Li atoms are removed from the Orth-C₁₄ monolayer. AIMD simulations conducted at 500 K reveals that the Orth-C₁₄ monolayer restores its initial configuration, demonstrating its remarkable stability. The bonding in single and double layers of Li atoms adsorbed on the Orth-C₁₄ monolayer is examined using the ELF in the subsequent analysis (Fig. S2). In this representation, the region between Orth-C₁₄ and Li has a value around 0.75, indicating that Orth-C₁₄ maintains structural stability even after adsorbing bilayer Li atoms, with strong covalent bonding persisting. The electronic orbital structure of the Orth-C₁₄ during bilayer Li atoms adsorption is systematically analyzed through PDOS calculation, manifesting significant contributions of the red C-py curve around the Fermi level. We find that the overlap between the Li-s orbital and C-2p orbitals suggests orbital hybridization between Li and the Orth-C₁₄ monolayer. As depicted in Fig. 8 (c-d), the overlap region is small, indicating that the hybridization degree of Li-s and C-2p orbitals is weak. This weak orbital interaction makes Li atoms more susceptible to undergo deintercalation, which effectively suppresses the volume expansion of the material during the cycling process.

The volume/area changes of electrode materials affect the cycling stability and safety of the battery. This work identifies the area change rate (ΔS) of Orth-C₁₄ monolayer the during Li atoms insertion and extraction processes. We determine the rate of area change rate (ΔS) by [54]:

$$\Delta S = \frac{S_{Li_xOrth-C_{14}} \pm S_{Orth-C_{14}}}{S_{Orth-C_{14}}} \times 100\% \quad (6)$$

Where $S_{Li_xOrth-C_{14}}$ and $S_{Orth-C_{14}}$ are the areas of the fully intercalated and the intrinsic structures, respectively. The lattice constants of the initial structure are $a = 17.50$ Å and $b = 8.70$ Å; after Li atom insertion, the lattice constants increase to 17.54 Å and 8.79 Å, respectively. A low area expansion rate of 1.3 % is calculated for Orth-C₁₄ which is lower

than those of penta-graphene (3.6 %), graphite (10 %), and bco-C₁₆ (13.4 %) [18,46,55]. The adsorption height between the Li atom and the Orth-C₁₄ substrate increases from 1.745 Å to 1.787 Å. This change reflects the strength of interaction between the Li atom and the substrate. Therefore, it is reasonably expected that Orth-C₁₄ as a possible anode ensures the excellent cycle stability of LIBs during the charge/discharge. Anodes with smaller expansion resist structural damage during cycling, improving longevity, energy density, and safety by mitigating short-circuit risks.

Achieving both high Li storage capacity and low diffusion barriers remains a significant challenge in the research of 2D materials. Although Me-C₈B₅ surpasses the Orth-C₁₄ monolayer in theoretical storage capacity (2846 mAh/g), its elevated diffusion barriers (0.66 eV) substantially limit rate capability [56]. Conversely, Θ-graphene achieves exceptional ionic conductivity (0.48 eV diffusion barrier) but suffers from inadequate storage capacity (877 mAh/g) [57]. Table 1 consolidates a variety of two-dimensional carbon materials and selected carbide electrode materials. By comparing key parameters such as the Li-ion diffusion barrier, theoretical storage capacity, and OCV, the overall competitiveness of Orth-C₁₄ is systematically evaluated. Furthermore, Orth-C₁₄ achieves an optimal balance of electrochemical properties, exhibiting not only high theoretical storage capacity and low diffusion barrier, but also moderate OCV. With its balanced electrochemical properties, Orth-C₁₄ shows great promise as an anode material for future batteries.

3.4. Comparative to Orth-C₁₄: an analysis with structurally analog of Orth-C₁₆

Orth-C₁₆ belongs to the family of sp²-bonded 2D carbon polymorphs with a lattice geometry similar to that of Orth-C₁₄, as reported in previous theoretical work. The structure consists primarily of heptagonal and octagonal rings in Fig. S3, and the lattice parameters are $a = 4.710$ Å and $b = 9.301$ Å. In Orth-C₁₆, the C—C bond length ranges from 1.39 to 1.46 Å, while the C = C bond length predominantly falls within 1.40

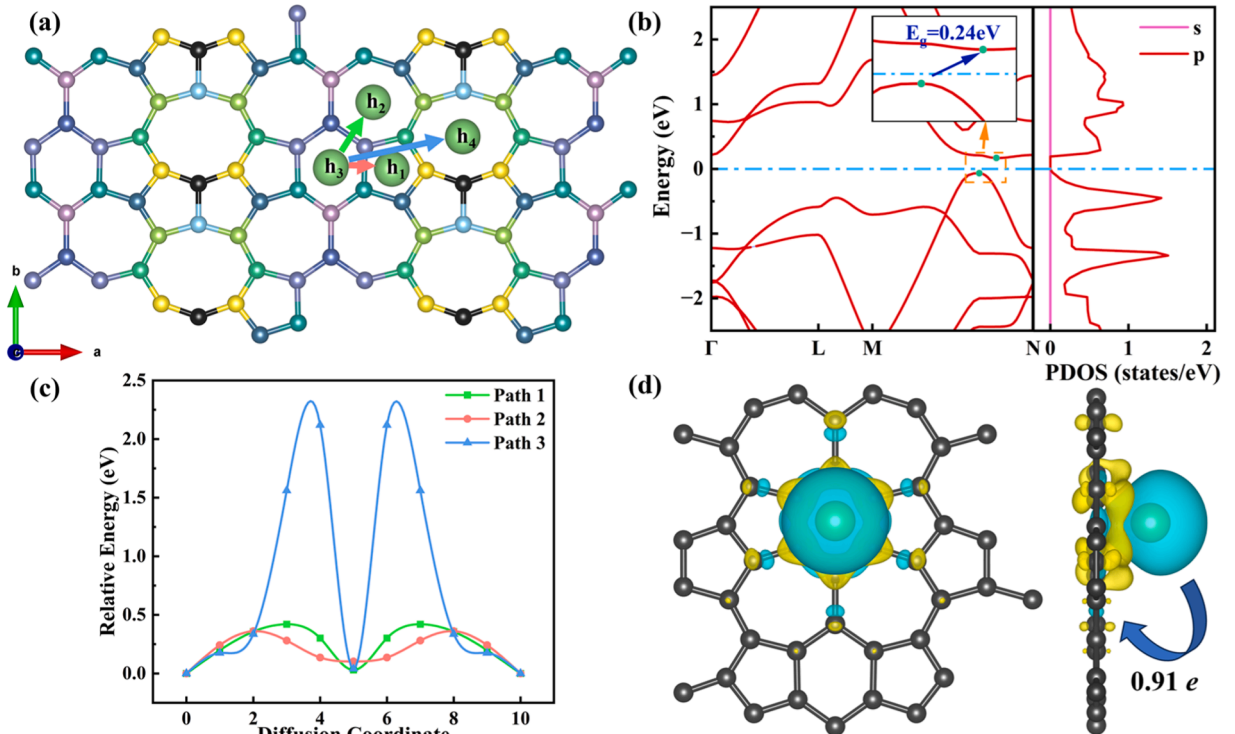


Fig. 9. (a) The schematic diagram of possible pathways for the diffusion of Li atom on the Orth-C₁₆ monolayer. (b) The band structure and PDOS of the Orth-C₁₆ monolayer. (c) The corresponding energy barrier profiles of three diffusion paths on images. (d) The CDD for the Li atom adsorption on the Orth-C₁₆ monolayer.

–1.47 Å, which is consistent with the previous results reported [21]. The potential of the Orth-C₁₆ monolayer as a LIB anode material is comprehensively quantified. Excellent dynamical and thermodynamic stability of the Orth-C₁₆ monolayer is confirmed by the absence of imaginary frequencies in its phonon dispersion and the stable energy evolution in AIMD simulations at 500 K for 5 ps, as shown in Fig. S4 (a-b). As depicted in Fig. 9(b), the electronic structure calculations identify it as a characteristic semiconductor, evidenced by its 0.24 eV direct band gap and vanishing DOS curve at the Fermi level.

With the intrinsic characteristics of the Orth-C₁₆ monolayer confirmed, the behavior of Li atoms is investigated, including a systematic calculation and comparison of their adsorption energies and positions at potential sites. (see Fig. S5). The results demonstrate that the *h* sites exhibit the most stable Li adsorption, particularly the *h*₃ site with an adsorption energy of –2.14 eV. Analysis of the Bader charge at the *h*₃ site reveals 0.91 *e* of charge transfer (see Fig. 9(d)). Calculate for Li adsorption at the *h* site of the Orth-C₁₆ monolayer identify three diffusion pathways with energy barriers of 0.41 eV, 0.36 eV, and 2.12 eV, respectively (see Fig. 9(c)). The relatively high diffusion barriers in the Orth-C₁₆ monolayer increase ionic transport resistance, impede electron conduction, and reduce the diffusion coefficient, consequently limiting the charge transfer efficiency of the electrode. In addition, we increase Li atoms (*x*) concentration in the Orth-C₁₆ monolayer from *x* = 3 (OCV = 0.570 V) to *x* = 12 (OCV = 0.120 V) and ultimately *x* = 24 (OCV = 0.057 V) (see Fig. S9). A theoretical storage capacity of 836.8 mAh/g is achieved by the material, supported by a maximum adsorption capacity of 12 Li atoms per side (24 Li atoms total). While both Orth-C₁₄ and Orth-C₁₆ are 2D carbon allotropes with polygonal carbon rings, there are differences in their properties. Orth-C₁₆ displays semi-conducting behavior with high Li diffusion barriers and modest theoretical storage capacity, whereas Orth-C₁₄ comprehensively demonstrates superior performance across the key evaluated metrics, making it the preferred lithium-ion battery anode.

4. Conclusions

In this work, we performed the first-principles calculations to investigate the structure, stability, adsorption, diffusivity, storage and electronic properties of the Orth-C₁₄ monolayer as a potential anode material for Li-ion batteries. The intrinsic stability of the Orth-C₁₄ monolayer is first confirmed through phonon spectra calculations, which show no imaginary frequencies and thus demonstrate its dynamical stability. Further validation comes from AIMD simulations at 500 K and 1000 K, where minimal total energy fluctuations and undistorted structure prove the excellent thermodynamic stability of the Orth-C₁₄ monolayer. The band structure and DOS diagrams manifest that Orth-C₁₄ has electronic bands crossing the Fermi level, indicating intrinsic metallicity dominated by the 2p orbitals. Regarding Li storage performance, the Orth-C₁₄ monolayer exhibits remarkable advantages, and its substantial Li adsorption energy of –1.78 eV at H1 sites ensures spontaneous and stable ion adsorption. The Bader charge analysis indicates a 0.97 *e* charge transfer from the Li atoms to Orth-C₁₄ monolayer. The energy barriers for three considered diffusion paths are calculated, revealing that the Orth-C₁₄ monolayer achieves an ultralow diffusion barrier of 0.12 eV, which significantly surpasses other 2D carbon allotropes. The Orth-C₁₄ exhibits an estimated high capacity of 956 mAh/g, positioning it among the few 2D carbon allotropes with metallic properties and high capacity. To simulate the charging/discharging process, we separately calculated the AIMD for the Orth-C₁₄ monolayer in both the fully adsorption and the desorption state. The Orth-C₁₄ monolayer showed maintained structural stability with minimal deformation and self-recovery tendency during the entire AIMD simulation. A minimal area changes of only 1.3 % is retained by its structure upon complete lithiation, manifesting its exceptional stability. These superior properties collectively enable the Orth-C₁₄ monolayer to achieve unprecedented performance in rate capability, cycling stability, and energy

density, positioning it as a next-generation anode material for advanced LIBs.

CRedit authorship contribution statement

Han Zhang: Writing – review & editing, Writing – original draft, Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Zilin Chen:** Writing – review & editing, Validation, Software, Methodology, Investigation, Data curation. **Huimin He:** Visualization, Investigation. **Qiuyan Lu:** Visualization, Investigation. **Huadong Zeng:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation. **Xinlu Cheng:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.surfin.2025.108228](https://doi.org/10.1016/j.surfin.2025.108228).

Data availability

Data will be made available on request.

References

- [1] J. Zhang, H. Huang, G. Zhang, Z. Dai, Y. Wen, L. Jiang, Cycle life studies of lithium-ion power batteries for electric vehicles: a review, *J. Energy Storage* 93 (2024) 112231.
- [2] O. Tasneem, H. Tasneem, X. Xian, Lithium-ion battery technologies for grid-scale renewable energy storage, *Next Res.* (2025) 100297.
- [3] T. Gao, Y. Han, D. Fraggadakis, S. Das, T. Zhou, C.-N. Yeh, S. Xu, W.C. Chueh, J. Li, M.Z. Bazant, Interplay of lithium intercalation and plating on a single graphite particle, *Joule* 5 (2021) 393–414.
- [4] A. Rajkamal, R. Thapa, Carbon allotropes as anode material for lithium-ion batteries, *Adv. Mater. Technol.* 4 (2019) 1900307.
- [5] Y. Wen, K. He, Y. Zhu, F. Han, Y. Xu, I. Matsuda, Y. Ishii, J. Cumings, C. Wang, Expanded graphite as superior anode for sodium-ion batteries, *Nat. Commun.* 5 (2014) 4033.
- [6] P. Patel, S. Patel, D. Chodvadiya, M.H. Dalsaniya, D. Kurzydowski, P.K. Jha, Two-dimensional α-SiX (X = N, P) monolayers as efficient anode material for Li-ion batteries: a first-principles study, *ACS Appl. Nano Mater.* 6 (2023) 2103–2115.
- [7] D. Chodvadiya, U. Jha, P. Śpiewak, K.J. Kurzydowski, P.K. Jha, Potential anodic application of 2D h-AIC for Li and Na-ions batteries, *Appl. Surf. Sci.* 593 (2022).
- [8] P. Patel, S. Patel, D. Chodvadiya, M.H. Dalsaniya, D. Kurzydowski, K. J. Kurzydowski, P.K. Jha, A density functional theory study on the assessment of α-CN and α-CP monolayers as anode material in Li-ion batteries, *J. Energy Storage* (2023) 71.
- [9] J. Ren, H. Wu, N. Zhang, J. Huo, The C4N/graphene heterostructure as a high-performance anode material for alkaline metal-ion batteries, *J. Alloys Compd.* 1010 (2025) 177773.
- [10] R.B. de Oliveira, D.D. Borges, L.D. Machado, Mechanical and gas adsorption properties of graphene and graphynes under biaxial strain, *Sci. Rep.* 12 (2022) 22393.
- [11] K.-S. Chen, I. Balla, N.S. Luu, M.C. Hersam, Emerging opportunities for two-dimensional materials in lithium-ion batteries, *ACS. Energy Lett.* 2 (2017) 2026–2034.
- [12] R. Rojaee, F. Shahbazian-Yassar, Two-dimensional materials to address the lithium battery challenges, *ACS Nano* 14 (2020) 2628–2658.

- [13] H.G. Ali, K. Khan, M.B. Hanif, M.Z. Khan, I. Hussain, M.S. Javed, H.A. Al-bonsrulah, M. Mosialek, M. Fichtner, M. Motola, Advancements in two-dimensional materials as anodes for lithium-ion batteries: exploring composition-structure-property relationships emerging trends, and future perspective, *J. Energy Storage* 73 (2023) 108980.
- [14] D.G. Ladha, A review on density functional theory-based study on two-dimensional materials used in batteries, *Mater. Today Chem.* 11 (2019) 94–111.
- [15] G. Huang, B. Cai, C. Zhan, P. Sun, Two-dimensional material as anode for lithium ion batteries: recent progress, *Int. J. Electrochem. Sci.* 15 (2020) 5416–5429.
- [16] Z. Yang, D. Zhang, K. Wang, J. He, J. Li, C. Huang, Investigating graphdiyne based materials for rechargeable batteries, *Nano Today* 46 (2022) 101588.
- [17] J. Feng, C. Yang, L. Zhang, F. Lai, L. Du, X. Yang, First-principle calculation of distorted T-carbon as a promising anode for Li-ion batteries with enhanced capacity, reversibility, and ion migration properties, *Carbon Energy* 2 (2020) 614–623.
- [18] B. Xiao, Y.-c. Li, X.-f. Yu, J.-b. Cheng, Penta-graphene: a promising anode material as the Li/Na-ion battery with both extremely high theoretical capacity and fast charge/discharge rate, *ACS Appl. Mater. Interfaces* 8 (2016) 35342–35352.
- [19] K.A.L. Lima, J.A.S. Laranjeira, N.F. Martins, A.C. Dias, J.R. Sambrano, D.S. Galvão, L.A.R. Junior, Petal-Graphyne: a novel 2D carbon allotrope for high-performance Li and Na ion storage, *J. Energy Storage* (2025) 130.
- [20] X.H. Cai, Q. Yang, S. Zheng, M. Wang, Net-C18: a predicted two-dimensional planar carbon allotrope and potential for an anode in lithium-ion battery, *Energy Env. Mater.* 4 (2021) 458–464.
- [21] H. Liu, Y. Yang, M. Xing, Q. Fan, Two-dimensional carbon allotrope with remarkable electron mobility and tunable band gap under uniaxial strain engineering, *Results Phys.* 58 (2024) 107435.
- [22] G. Kresse, J. Furthmüller, Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set, *Comput. Mater. Sci.* 6 (1996) 15–50.
- [23] G. Kresse, J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set, *Phys. Rev. B* 54 (1996) 11169.
- [24] P.E. Blöchl, Projector augmented-wave method, *Phys. Rev. B* 50 (1994) 17953.
- [25] J.P. Perdew, K. Burke, M. Ernzerhof, Generalized gradient approximation made simple, *Phys. Rev. Lett.* 77 (1996) 3865.
- [26] S. Grimme, J. Antony, S. Ehrlich, H. Krieg, A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H-Pu, *J. Chem. Phys.* (2010) 132.
- [27] S. Grimme, Semiempirical GGA-type density functional constructed with a long-range dispersion correction, *J. Comput. Chem* 27 (2006) 1787–1799.
- [28] S. Baroni, S. De Gironcoli, A. Dal Corso, P. Giannozzi, Phonons and related crystal properties from density-functional perturbation theory, *Rev. Mod. Phys.* 73 (2001) 515.
- [29] V. Wang, N. Xu, J.-C. Liu, G. Tang, W.-T. Geng, VASPKIT: a user-friendly interface facilitating high-throughput computing and analysis using VASP code, *Comput. Phys. Commun.* 267 (2021) 108033.
- [30] J. Rong, H. Dong, J. Feng, X. Wang, Y. Zhang, X. Yu, Z. Zhan, Planar metallic carbon allotrope from graphene-like nanoribbons, *Carbon* N. Y. 135 (2018) 21–28.
- [31] B. Andrea, S. Francesco, C. Mattia, N. Louis, C. Maurizio, S. Mauro, Molecules-Oligomers-Nanowires-Graphene nanoribbons: a bottom-up stepwise on-surface covalent synthesis preserving long-range order, (2015).
- [32] Q. Fan, L. Yan, M.W. Tripp, O. Krejčí, S. Dimosthenous, S.R. Kachel, M. Chen, A. S. Foster, U. Koert, P. Liljeroth, Biphenylene network: a nonbenzenoid carbon allotrope, *Science* 372 (2021) 852–856.
- [33] F. Liu, P. Ming, J. Li, Ab initio calculation of ideal strength and phonon instability of graphene under tension, *Phys. Rev. B—Condens. Matter Mater. Phys.* 76 (2007) 064120.
- [34] A.D. Becke, K.E. Edgecombe, A simple measure of electron localization in atomic and molecular systems, *J. Chem. Phys.* 92 (1990) 5397–5403.
- [35] B. Ball, C. Chakravarty, P. Sarkar, Two-dimensional covalent triazine framework as a promising anode material for Li-ion batteries, *J. Phys. Chem. C* 123 (2019) 30155–30164.
- [36] A. Ghosh, S. Mandal, Sarkar, 2D homogeneous holey carbon nitride: an efficient anode material for Li-ion batteries with ultrahigh capacity, *Chemphyschem* 23 (2022) e202200182.
- [37] E. Santos, K. Lima, L.Ribeiro Junior, Proposing TODD-graphene as a novel porous 2D carbon allotrope designed for superior lithium-ion battery efficiency, *Sci. Rep* 14 (2024) 6202.
- [38] N.F. Martins, J.A. Laranjeira, K.A. Lima, L.A. Cabral, L. Junior, J.R. Sambrano, HOP-graphene: a high-capacity anode for Li/Na-ion batteries unveiled by first-principles calculations, *arXiv preprint arXiv:2505.04804*, (2025).
- [39] S. Wang, B. Yang, H. Chen, E. Ruckenstein, Popgraphene: a new 2D planar carbon allotrope composed of 5-8-5 carbon rings for high-performance lithium-ion battery anodes from bottom-up programming, *J. Mater. Chem. A* 6 (2018) 6815–6821.
- [40] K. Lu, T. Wang, X. Li, L. Dai, J. Yin, Y. Ding, A new 2D carbon allotrope C568 as a high-capacity electrode material for lithium-ion batteries, *Fuller. Nanotub. Carbon Nanostructures* 30 (2022) 385–391.
- [41] E. Santos, K. Lima, F. Mendonça, D.d. Silva, W. Giazza, L.R. Junior, PHOTH-graphene: a new 2D carbon allotrope with low barriers for Li-ion mobility, *Sci. Rep.* 14 (2024) 9526.
- [42] L. Lu, R. Gallenstein, X. Liu, Y. Lin, S. Lin, Z. Chen, Holey penta-hexagonal graphene: a promising anode material for Li-ion batteries, *PCCP* 26 (2024) 7335–7342.
- [43] X. Xiong, H.-B. Cao, Z. Lu, C.-S. Liu, X.-J. Ye, Theoretical prediction on irida-graphene monolayer as promising anode material for lithium-ion batteries, *Comput. Mater. Sci* 244 (2024) 113225.
- [44] L. Shao, X. Duan, Y. Li, F. Zeng, H. Ye, C. Su, P. Ding, Two-dimensional planar BGe monolayer as an anode material for sodium-ion batteries, *ACS Appl. Mater. Interfaces* 13 (2021) 29764–29769.
- [45] A. Majid, U. Najam, S. Ahmad, M. Alkhdher, On the prospects of using B4C3 as a potential electrode material for lithium-ion batteries, *Mater. Sci. Semicond. Process.* 176 (2024) 108320.
- [46] D. Fan, A.A. Golov, A.A. Kabanov, C. Chen, S. Lu, X. Li, M. Jiang, X. Hu, A new sp²-sp³-hybridized metallic carbon network for lithium-ion battery anode with enhanced safety and lithium-ion diffusion rate, *J. Phys. Chem. C* 123 (2019) 15412–15418.
- [47] C. Kou, Y. Tian, M. Zhang, E. Zurek, X. Qu, X. Wang, K. Yin, Y. Yan, L. Gao, M. Lu, M-graphene: a metastable two-dimensional carbon allotrope, *2D Mater.* 7 (2020) 025047.
- [48] S. Wang, Y. Si, B. Yang, E. Ruckenstein, H. Chen, Two-dimensional carbon-based auxetic materials for broad-spectrum metal-ion battery anodes, *J. Phys. Chem. Lett.* 10 (2019) 3269–3275.
- [49] Z. Liu, Y. Ma, Y. Niu, Y. Zhao, Y. Zhang, H. Hou, K. Meng, X. Yu, ZW-21: a novel two-dimensional graphene-based anode material for the lithium-ion battery, *Diam. Relat. Mater.* 128 (2022) 109298.
- [50] Y.-T. Yan, Z.-G. Shao, C.-L. Wang, L. Yang, Planar carbon allotrope B-graphyne as lithium-ion battery anode materials, *Chem. Phys. Lett.* 804 (2022) 139897.
- [51] W. Wang, J. Meng, Y. Hu, J. Wang, Q. Li, J. Yang, Thgraphene: a novel two-dimensional carbon allotrope as a potential multifunctional material for electrochemical water splitting and potassium-ion batteries, *J. Mater. Chem. A* 10 (2022) 9848–9857.
- [52] X. Wang, Z. Feng, J. Rong, Y. Zhang, Y. Zhong, J. Feng, X. Yu, Z. Zhan, Planar net- τ : a new high-performance metallic carbon anode material for lithium-ion batteries, *Carbon* N. Y. 142 (2019) 438–444.
- [53] Z. Cheng, X. Zhang, H. Zhang, J. Gao, H. Liu, X. Yu, X. Dai, G. Liu, G. Chen, Two-dimensional metallic carbon allotrope with multiple rings for ion batteries, *PCCP* 23 (2021) 18770–18776.
- [54] S.-C. Mo, X.-Y. Qiu, J. Nie, G.-Y. Li, F. Ning, S.-Z. Chen, Long-range ordered porous carbon C52: a high-performance anode metallic material for next-generation Li-ion batteries with extremely high capacity, *J. Energy Storage* 110 (2025) 115286.
- [55] J. Liu, S. Wang, Q. Sun, All-carbon-based porous topological semimetal for Li-ion battery anode material, *Proc. Natl. Acad. Sci.* 114 (2017) 651–656.
- [56] R. Chen, Y. Gong, Z. Chen, Z. Zhu, D. Chen, M. Cheng, Me-C8B5: a two-dimensional carbon boride monolayer with potential applications as anode material for Li-ion battery, *J. Alloys Compd.* 1020 (2025) 179324.
- [57] S. Wang, B. Yang, H. Chen, E. Ruckenstein, Reconfiguring graphene for high-performance metal-ion battery anodes, *Energy Storage Mater.* 16 (2019) 619–624.
- [58] M. Sajjad, K. Badawy, J.A. Larsson, R. Umer, N. Singh, Two dimensional holey graphyne: an excellent anode and anchoring material for metal-ion and metal-sulfur batteries, *Carbon* N. Y. (2023) 214.
- [59] J. Gomez Quispe, B. Ipaves, D.S. Galvao, P.A.d.S. Autreto, TPDH-graphene as a new anodic material for lithium ion battery: dFT-based investigations, *ACS Omega* 9 (2024) 39195–39201.